

SUMAN GHOSH

Lead Applied AI Scientist | AI Systems Architect | Production ML

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PROFILE

Lead Applied AI Scientist with 12+ years of experience delivering production AI systems across Banking, Healthcare, and Manufacturing. Currently leading AI transformation initiatives at HSBC as Senior AVP, architecting agentic systems, transformer-based prediction models, and enterprise LLM platforms. Expertise spans multi-agent orchestration, production RAG, LLMops, fraud analytics, AI platform engineering, and scalable ML infrastructure, underpinned by a master's in economics from Delhi School of Economics. Recognised as HSBC AI Advanced Ambassador, open-source contributor, and technical author

EXPERIENCE

Senior AVP — AI Transformation · HSBC

Jan 2022 - Present Bengaluru

Lead AI transformation initiatives across HSBC IWPB by delivering production grade agentic systems, transformer-based architectures, and LLM powered platforms

Autonomous Fraud Intelligence Platform

- Architected a closed-loop agentic system — Monitor → Feature Discovery → Retrain — that autonomously detects model drift, proposes novel features via LLM reasoning, and rebuilds challenger models without manual intervention.
- Reduced model redevelopment cycle by **68%**; real-time monitoring dashboard deployed across fraud operations and call centres to cut customer friction.
- Orchestrated via graph-based sequential and loop agents (MAF) with modular design, structured logging, and a full observability layer.

Enterprise Multi-Agent KYC Onboarding Platform

- Designed an end-to-end multi-agent KYC onboarding platform supporting multi-market regulatory workflows. The orchestration pipeline automates document extraction, classification, SOP rule validation, and decisioning, generating structured verification reports with confidence scores and complete audit trails.
- Engineered the backend using FastAPI and developed the frontend using React.js.
- Delivered in partnership with Microsoft as a reference architecture for regulated-industry KYC automation; reducing manual review effort by **84%** and generating an estimated annual cost saving of **\$1.2M**.

Next Best Action Intelligence Engine

- Designed a next-generation recommendation engine to improve relationship manager decision-making by combining customer behaviour with product relationship intelligence.
- Leveraged Knowledge Graph (TransE embeddings over product–customer relationships) with a transformer encoder via cross-attention - enabling simultaneous reasoning over sequential behaviour and structured relational context.
- Validated the architecture on **2M+ customer** interactions, demonstrating a **16%** improvement in Recall@3 over the baseline recommendation model.

Enterprise Knowledge Intelligence Platform

- Architected an enterprise AI platform enabling Finance users to retrieve governed organisational knowledge through LLM-powered semantic search.
- Implemented production-grade RAG capabilities including multi-query, hybrid retrieval & reranking, LLMops instrumentation, AI safety guardrails, and grounding controls, deployed through Azure Web Apps for enterprise process intelligence.

Enterprise ML Platform (NxtGen)

- Spearheaded a no-code **enterprise ML platform** for finance analytics; reduced model go-live timelines by **40%**.
- Aligned with independent validation on architecture; provides an interactive candidate model selection interface with transparent audit trails for business stakeholders.

Financial Assurance Intelligence Platform

- Established a layered financial assurance platform combining deterministic risk rules with unsupervised anomaly detection (Isolation Forest + Autoencoder) to identify previously unseen ledger anomalies.
- Adopted for monthly analytic review, surfacing **10+ unusual cases** per cycle; enables proactive assurance gap remediation.

Treasury Hedge Optimization Engine

- Prepayment prediction model (logistic regression + neural network parameter optimisation) enabling treasury team to optimise the hedge portfolio by **\$3.2B**.
- Production Python codebase following TDD principles; fully integrated with IRRBB stress testing workflows.

Senior Data Scientist - Manager · Citi

Jul 2018 – Jan 2022 · Bengaluru

- Account-level campaign targeting (XGBoost + rebalancing + Logistic Regression); each campaign delivered **\$13.2M** additional sales. Annual portfolio impact of **\$78M**. Successfully passed Model Risk Management, Fair Lending, and implementation reviews.
- Covid-19 impact analysis: multi-class driver model (Random Forest) with rule-based arrest triggers deployed directly into campaign logic.

Senior Lead - Advanced Analytics · GE Healthcare

Aug 2015 - Jul 2018 · Bengaluru

- B2B account targeting ecosystem (propensity models + Markov Chain timing); generated **\$600K new orders** in Africa and \$2M in pipeline for the value CT segment.
- Quarterly market revenue forecasting engine on GE Cloud, enriched with NLP topic modelling on GDELT government sentiment signals; used in quarterly op-mech reviews.

Senior Business Analyst · TCS R&D

Jul 2013 - Jul 2015 · Bengaluru

- Social media analytics platform (Twitter, Facebook, YouTube): LDA topic extraction, association analysis, sentiment analysis for client product engagement strategy.

OPEN SOURCE & PUBLICATIONS

- ContextDriftAnalyzer — open-source Python package for detecting context drift in LLM conversations; published on PyPI.
- Technical articles on Agentic AI, LLM inference, transformer architectures, and production AI systems, published on Medium and shared with the AI community via LinkedIn.

Medium medium.com/@whitemansion1990

GitHub github.com/Suman-Git-DS

TECHNICAL SKILLS

- **AI & Transformer Systems**: Transformer architectures, Knowledge Graphs, Predictive Modelling, NLP, Anomaly Detection, Time Series Forecasting, API development
- **Agentic & LLMOps**: Multi-agent orchestration, Microsoft Agent Framework, LangChain/Langgraph, Production RAG, AI safety guardrails, LLMops instrumentation
- **MLOps & Infrastructure**: MLflow, Evidently, Prometheus, Grafana, Docker | Containerization, FastAPI, WebApp deployment
- **Languages**: Python | PySpark | R | SQL (BigQuery)
- **Cloud** - Microsoft Azure | Google Cloud Platform

RECOGNITION & LEADERSHIP

- **HSBC AI Advanced Ambassador** — recognised for driving enterprise AI adoption, mentoring teams on agentic and LLM system design, and shaping the bank's AI transformation roadmap.
- **Cross-functional delivery leader**: aligned Model Risk Management, Fair Lending, Independent Validation, and senior business stakeholders across production AI programmes in regulated environments.